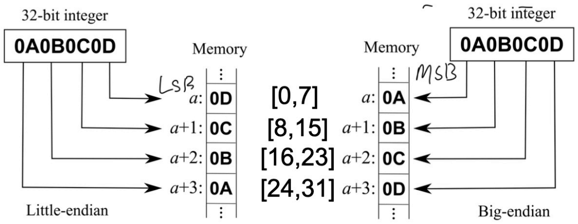
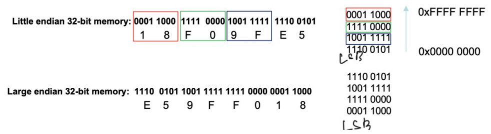
## Endianness

Endian refers to the order of **bytes** stored in the memory or transmitted over digital links

Big endian: The most significant **byte** is stored first, at the lowest memory address
Little endian: The least significant **byte** is stored first, at the lowest memory address.

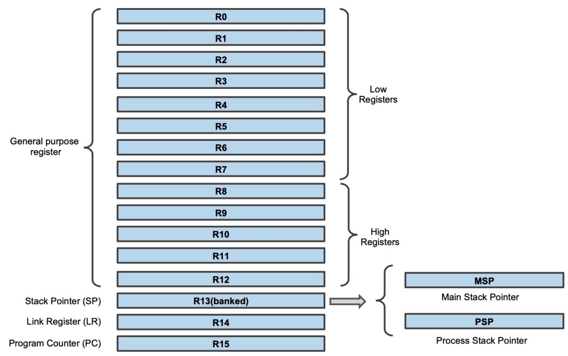



Home work: how is 0x18F09FE5 saved in both little and big endian formats?

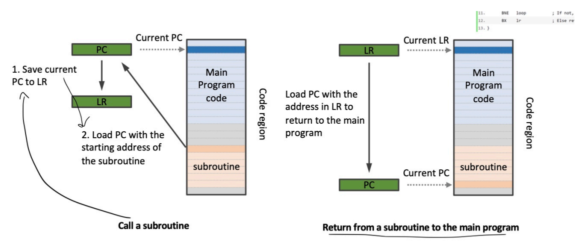



## Cortex – M0 Registers
Register bank

Special registers


R0 – R12: general purpose registers
Low registers (R0 – R7)
High registers (R8 – R12)

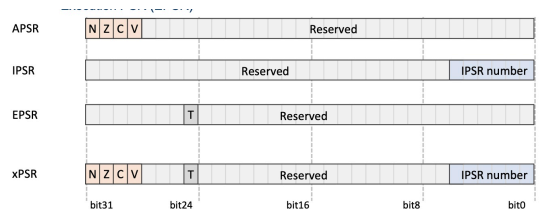



R13: Stack Pointer (SP):
Records the current address of the stack.
Used for saving the context of a program while switching between tasks

R14: Link Register (LR):
The LR is used to store the return address of a subroutine or a function call;

R15: Program Counter (PC):
Records the address of the current instruction code;

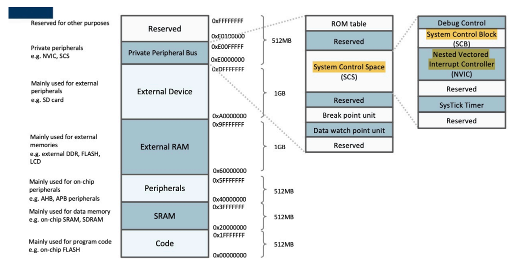
xPSR, combined Program Status Register




## Cortex M0 Memory Map

totally 4 GB

Code Region:
• Primarily used to store program code;
• Can also be used for data memory;
• On-chip memory, such as on-chip FLASH.




SRAM Region
• Primarily used to store data, such as heaps and stacks;
• Can also be used for program code;
• On-chip memory;

Peripheral Region
• Primarily used for peripherals, such as Advanced High-performance Bus (AHB) or Advanced Peripheral Bus (APB) peripherals;
• On-chip peripherals.

External RAM Region
• Primarily used to store large data blocks, or memory caches;
• Off-chip memory, slower than on-chip SRAM region.

External Device Region
• Primarily used to map to external devices;
• Off-chip devices, such as SD card.

Internal Private Peripheral Bus (PPB)
• Used inside the processor for processor's internal control;